PERFORMING RIGS

AutoPilot Firmware Validation Report

Document	AP-VVR-004
Revision	Rev A
Product	AutoPilot
Hardware Revision	Hardware Revision 2 (Batch-2 production receipt, PCB silk 52_P0_V01R04)
Firmware Revision	Autopilot_firm05-08 · 915 MHz · UI AutoPilot V01 R39 · Remote V01 R42
Validation Date	September 1, 2026
Prepared By	Nathan Destro, Performing Rigs
Location	Bench / shop

Validation Summary

Validation Status	**CLOSED**
Functional Result	**FAIL**
Gate A (F-001 on Rev 2)	**FAIL**

Validation Status — Validation is **closed** for this receipt. Intentional dry-contact recognition on Rev 2 is strong, and dedicated idle watches (T-01–T-03) passed when Chain B wiring was not connected. However, **Gate A FAIL**: Rev 2 reproduces the **AP-VVR-003 F-001 hazard class** on **Chain B-1 and Chain B-2** (see **Test wiring reference**). A Rev 1 cross-check shows **the same PASS/FAIL split** — the Rev 2 trigger rework did not fix Chain B. Batch-2 units must remain **quarantined**.

Functional Result — Intentional dry-contact fires met targets (**15+**, zero misses). Chain A idle-quiet on Rev 1 and Rev 2. Chain B false-trigger behavior is a **critical FAIL**. These results **do not** constitute Batch-2 engineering release approval.

Purpose. Accept or reject the **Rev 2 production batch** for engineering release using the **same wiring faults documented on Rev 1** in AP-VVR-003. Receipt of hardware is **not** engineering acceptance.

Unit under test	Serial / ID	Notes
UUT-1	No serial numbers	Primary — full matrix
UUT-2	No serial numbers	F-001 spot-check — same three chains as UUT-1

Chronology (traceability). AP-VVR-003 sent 2026-08-06 (F-001 OPEN, Rev 1 only). No technical reply to Q-001 / Q-002 / Q-003a / Q-005 before ~2026-08-24 production shipment (no advance shipping notice). AP-VVR-004 bench session 2026-09-01.

1. Executive Summary

Hardware Revision 2 unit(s) from the August 2026 production receipt (~15 assemblies) were validated against AP-VVR-003 carry-forward items, with **centerpiece** reproduction of **AP-VVR-003 F-001** on studio-realistic wiring. Two units (UUT-1 full matrix, UUT-2 F-001 spot-check) showed the same result. A **Hardware Revision 1** bench unit was cross-checked on the **same three wiring paths** (see Test wiring reference).

Positive results. Cold-start, soft-reset, and power-cycle idle watches (T-01–T-03) passed with **no** spontaneous false triggers when Chain B wiring was not connected. **Chain A** passed on Rev 1 and Rev 2 (~1667 mV / ~3300 mV INFO readout; quiet). Intentional dry-contact fires: **15+ consecutive, zero misses**; INFO mV steps directly **3300 0** with no intermediate values. Rev 2 trigger menu per Option C confirmed as designed.

Critical concern — Gate A FAIL. Both **Chain B** wirings produce **continuous false triggering** on **Rev 1 and Rev 2**:

- **Chain B-1 — Spark+ output:** AP **|** Pilot Input Adapter **|** 25' 4-Pin XLR Cable **|** Spark+ output
- **Chain B-2 — idle bench supply:** AP **|** Pilot Input Adapter **|** 25' 4-Pin XLR Cable **|** Wire Lead **|** bench supply, not commanding

Rev 2 INFO on Chain B-1 reads **0 mV** (closed dry contact). Rev 1 Chain B-2 read **1925–1954 mV** in 12/24 Vdc mode during AP-VVR-003-class bench work. The Rev 2 trigger-circuit rework **did not eliminate** the Chain B hazard class.

Process / scope. Batch-2 arrived ~2026-08-24 **without** advance shipping notice, before AP-VVR-003 (2026-08-06) was answered and before Rev 2 Gate A evidence existed. Receipt is **not** engineering acceptance; batch remains **quarantined**.

2. Evidence Summary

Validation Item	Result
T-01 Cold power-on idle (false-trigger watch)	**PASS** (Chain A only; Chain B not connected)
T-02 Soft reset / reboot idle watch	**PASS**
T-03 Full power-cycle idle watch	**PASS**
T-04 Idle with nearby motion / noise	**NOT TESTED**
T-05 Relay / dry-contact intentional fires	**PASS** (15+, 0 misses)
T-06 12 V intentional fires	**NOT TESTED** (N/A Rev 2 — Option C)
T-07 24 V intentional fires	**NOT TESTED** (N/A Rev 2 — Option C)
T-08 9 V / mode cross-check	**NOT TESTED** (N/A Rev 2; Rev 1 Chain B-2 documented)
T-09 Monitoring / INFO during signal	**FAIL** — RUNNING UI desync on Chain B
T-10 Startup / minimum current	**NOT TESTED**
T-11 HW Rev 2 identification (silk / photos)	**PASS**
T-12 Rev 1 vs Rev 2 same wiring	**FAIL** — Chain B on both revs; Chain A quiet on both
T-13 Long idle soak	**NOT TESTED**
Gate A — Chain B false trigger	**FAIL** — see F-001 and wiring reference
Chain A (see wiring reference)	**PASS** (quiet; Rev 1 + Rev 2; UUT-1 + UUT-2)
Chain B-1 / B-2 (see wiring reference)	**FAIL** — continuous false triggering (Rev 1 + Rev 2)
RF channel (remote only; ch 1–6 in-session)	**PASS** in-session; **FAIL** cold-boot pairing unless remote on ch 1

Test wiring reference

All paths start at **AP Trigger Input**. Complete chains below — no truncation.

Test path	Rev 1	Rev 2
Chain A — Spark+ input: AP **|** Pilot Input Adapter **|** 25' 4-Pin XLR Cable **|** Relay Adapter **|** Spark+ input	**PASS** — quiet	**PASS** — quiet
Chain B-1 — Spark+ output: AP **|** Pilot Input Adapter **|** 25' 4-Pin XLR Cable **|** Spark+ output	**FAIL** — continuous false triggering	**FAIL** — continuous false triggering
Chain B-2 — idle bench supply: AP **|** Pilot Input Adapter **|** 25' 4-Pin XLR Cable **|** Wire Lead **|** bench supply, not commanding	**FAIL** — continuous false triggering	**FAIL** — continuous false triggering

Trigger-input voltage observations (INFO / monitor)

Condition	Approximate readout
Standby / open (mV)	Settings trigger voltage: 0 mV (0 shown red). INFO Chain A: ~3300 mV Rev 2 / ~1667 mV Rev 1; ~74 mV flickers ~every 21–27 s on Rev 2 baseline.
Dry contact active (mV)	Intentional closure: 3300 mV 0 mV step only; no intermediate mV values observed.
F-001 fault setup — supply connected, not activated (mV)	**Chain A:** AP **|** Pilot Input Adapter **|** 25' 4-Pin XLR Cable **|** Relay Adapter **|** Spark+ input — ~1667 mV Rev 1 / ~3300 mV Rev 2, quiet. **Chain B-1:** AP **|** Pilot Input Adapter **|** 25' 4-Pin XLR Cable **|** Spark+ output — 0 mV Rev 2, continuous false triggering (Rev 1 same). **Chain B-2:** AP **|** Pilot Input Adapter **|** 25' 4-Pin XLR Cable **|** Wire Lead **|** bench supply, not commanding — continuous false triggering Rev 1 and Rev 2 (Rev 1: 1925–1954 mV in 12/24 V mode).
Polarity swap — Chain B-1 idle INFO (mV)	3300 mV — quiet (Spark+ powered and unpowered)
Polarity swap — Chain B-2 idle INFO (mV)	First connect: AP activates once; then 3300 mV quiet. 0/5/9 V no trip; 10 V 0 mV + trigger
Bench supply: output OFF, supply powered (V / at terminals)	-0.125 V
Bench supply: output ON @ 0 V (V at terminals)	0.04 V
Bench supply: output ON @ 12 V (V at terminals)	12.08 V

Rev 2 baseline note: Standby Chain A ~3300 mV (vs Rev 1 ~1667 mV). Periodic ~74 mV flickers on 3300 mV baseline every ~21–27 s observed with no motion during watch.

3. Findings

F-001 — Continuous false trigger on Chain B (Gate A — Rev 2 reproduces AP-VVR-003 class)

On **Chain B** studio hazard wirings, AutoPilot **continuously false-triggered** programmed moves (M3 in this session) on **Hardware Revision 2** (UUT-1 and UUT-2) and on **Hardware Revision 1** cross-check. Complete wiring paths are listed in the **Test wiring reference** table above.

Chain A (PASS — Rev 1 and Rev 2)

Chain A — Spark+ input: AP **|** Pilot Input Adapter **|** 25' 4-Pin XLR Cable **|** Relay Adapter **|** Spark+ input

PASS — quiet. INFO ~1667 mV Rev 1 / ~3300 mV Rev 2.

Chain B (FAIL — Rev 1 and Rev 2)

Chain B-1 — Spark+ output: AP **|** Pilot Input Adapter **|** 25' 4-Pin XLR Cable **|** Spark+ output

FAIL — continuous false triggering on Rev 1 and Rev 2. Rev 2 INFO **0 mV** (closed dry contact).

Chain B-2 — idle bench supply: AP **|** Pilot Input Adapter **|** 25' 4-Pin XLR Cable **|** Wire Lead **|** bench supply, not commanding

FAIL — continuous false triggering on Rev 1 and Rev 2. Rev 1 INFO **1925–1954 mV** in 12/24 Vdc trigger mode (AP-VVR-003 F-001 class).

Session notes: UUT-1 and UUT-2 showed the same results as the table above. An extended studio variant of Chain B-1 (25 ft XLR, Wire Lead, Relay Adapter before Spark+ output) also failed — same class as B-1. Cleared only by disconnecting Spark+ end or bench wire-lead end; upstream chain to AP could remain connected and quiet.

Engineering Impact

- Unexpected motion from Chain B wiring is a **critical** production hazard on studio sets (Spark+ output relay path, idle bench supply).
- Rev 2 trigger-circuit rework changed Chain A baseline mV but **did not fix** Chain B on Spark+ output; Rev 1 Spark+ output fails the same way.
- INFO readout on Chain B (**0 mV** on Spark+ output Rev 2) does **not** indicate a safe/open condition for operators.

EMS: AP-001 (related / expanded)

Severity: Critical

Status: OPEN — **Gate A FAIL**

F-002 — RUNNING UI desync during Chain B fault

During Chain B continuous-false-trigger conditions (wiring paths in **Test wiring reference**), exiting settings/monitoring (moves disabled) caused motor motion once triggers re-enabled (operator-expected). The remote **did not show RUNNING** during that motion. Rev 2: navigating to M3 caused unexpected return-to-A; UI desync briefly paused the continuous loop until the move was activated on the UI. Rev 1 (Chain B-1 / B-2): move did not run immediately on INFO exit; re-running the programmed move restarted repeated triggering. Observed on **Rev 1 and Rev 2**.

Severity: Medium (operator awareness / diagnostics)

Status: OPEN

*(Positive finding **F-003** — intentional dry-contact PASS — appears at the end of this report, before Revision History.)*

F-004 — Rev 2 trigger menu (Option C) confirmed

CONFIRMED by design (Option C): Rev 2 no Trigger threshold/mode line; Rev 1 keeps Relay Contact / 12/24 Vdc selector.

Severity: Info

Status: CONFIRMED by design

F-005 — Batch shipped before Rev 2 validation / unanswered AP-VVR-003

Production batch received ~2026-08-24 before Rev 2 engineering validation and before Noxon answered AP-VVR-003 technical questions. No advance shipping notice.

Severity: Info (process)

Status: Documented

F-006 — RF channel cold-boot pairing

RF channel is set on the **remote only** (no AP channel switch). Channels 1–6 operate correctly in-session when changed on the remote. After AP power-off/reboot, link does **not** restore until the remote is returned to **RF channel 1** (remote may still display ch 2–6 from the prior session).

Severity: Low / Medium (field usability)

Status: OPEN

4. Questions for Noxon

Q-001

Gate A **FAIL**: What is the root cause of **Chain B** continuous false triggers on Rev 2, and why did the Rev 2 trigger redesign not fix the Spark+ **output** path that also fails on Rev 1?

Q-002

State disposition for **all Rev 1 boards** — sold units and Performing Rigs inventory — given Rev 1 reproduces Chain B (Spark+ output and idle bench supply) and Rev 2 Gate A **FAIL**.

Q-003

Why was Batch-2 shipped (~2026-08-24) before AP-VVR-003 was answered and before Performing Rigs Rev 2 Gate A evidence existed? What is Noxon's required validation gate before shipment?

Q-004

What minimum factory / sample validation did Noxon perform on this production lot?

Q-005

RF channel: is cold-boot pairing to channel 1 only intended? What is the operator procedure for channels 2–6 after power-cycle?

Q-006

Confirm advance shipping notice / packing list / tracking before units leave Noxon (process).

5. Recommendations

R-001

Treat **F-001 / Gate A FAIL** as a **critical** open issue. **Do not sell** from the August 2026 Batch-2 receipt.

R-002

Keep Batch-2 **quarantined** until Noxon provides root cause, corrective action, and written disposition.

R-003

Send this report (wiring table, voltage observations, Rev 1 comparison) to Iker / Noxon with emphasis on Q-001 / Q-002 / Q-003.

R-004

Do not grant engineering release for **Hardware Revision 2** based on Rev 1 bench results or Noxon factory claims until Chain B is resolved and verified on Performing Rigs bench.

R-005

Document interim operator guidance: treat **Chain B** wiring paths (see **Test wiring reference**) as **unsafe** until Noxon disposition. **Chain A** was quiet in this session but is not a substitute for resolving Chain B.

6. Remaining Validation

- Noxon response and corrective-action verification on Chain B (both B-1 and B-2)
- Rev 2 re-test after Noxon fix — Gate A must **PASS** on UUT-1 + spot-check UUT-2
- Rev 1 unified field-support policy (firmware vs board replacement vs customer advisory)
- T-04 studio noise / nearby motion if needed for AP-001 closure
- T-10 startup current measurement (instrumentation)

Engineering Assessment

Dimension	Conclusion
Gate A — F-001 Chain B on Rev 2	**FAIL**
Chain A — Spark+ input (Rev 1 + Rev 2)	**PASS** (quiet this session)
Intentional dry-contact fires (Rev 2)	**PASS**
Idle spontaneous false-trigger watch (T-01–T-03)	**PASS** (Chain B not connected)
Diagnostics / RUNNING UI (Chain B)	**FAIL** (F-002)
Hardware Revision 2 Batch-2 release	**NOT CLEARED**
Rev 1 field disposition	**Required** — Noxon Q-002
Overall Validation Status	**CLOSED**
Overall Functional Result	**FAIL**

Shipment posture: Do not sell from this batch. Engineering approval to ship Batch-2 requires (1) Noxon root cause and corrective action for Chain B, (2) Performing Rigs re-validation Gate A **PASS**, and (3) written Rev 1 field disposition.

F-003 — Intentional dry-contact recognition met targets (Rev 2)

Dry contact intentional fires: 15+ consecutive, zero misses. INFO mV steps 3300 0 only.

Mode	Count	Misses
Dry contact (intentional)	15+	0

Severity: Info

Status: Validation Passed (intentional-fire scope)

Appendix — Noxon Response / Retest (Sep 2026)

Noxon reply (Iker Apalategi, 2026-09-02): Root cause attributed to idle connected sources presenting 0 V to the AP trigger input, which the input interprets identically to a dry relay closure. Rev 2 is designed to detect relay contact (0 V), 12 V, and 24 V; no inactive state exists for 0 V idle push-pull outputs. Noxon factory test used bench/laptop supplies with one wire always connected and the other manually connected/disconnected — this topology never holds an idle source on the line, explaining why Noxon did not reproduce Chain B. Two proposed mitigations: (1) **polarity swap** — GND/- to TRS tip, + to sleeve; (2) **680 series resistor** in one XLR conductor — compatible with 12/24 V sources only, not dry-contact cables. RF cold-boot behavior confirmed expected when powered via PSU instead of built-in switch; EEPROM-on-set fix planned. UI/INFO exit behavior to be reviewed.

Retest item	Result	Notes
VT-014 Chain B-1 polarity swap idle	**PASS**	2026-09-03: B-1 Spark+ output idle 3300 mV, quiet ≥10 min; quiet with Spark+ powered; button fire OK.
VT-015 Chain B-2 polarity swap idle	**FAIL**	2026-09-03: sustained idle PASS (3300 mV); connection transient FAIL — first XLR connect fires AP once (repeatable). Command trip ~10 V. See AP-VVR-005 F-002.
VT-016 Chain A + dry-c ontact still OK	**PASS**	Chain A unpowered Spark+ still 3300 quiet. Dry-contact 10/10 via Wire Lead path (not Chain A Spark+ input).

Polarity retest voltage observations:

Condition	Reading
Polarity swap — Chain B-1 idle INFO (mV)	3300 mV — quiet (Spark+ powered and unpowered)
Polarity swap — Chain B-2 idle INFO (mV)	First connect: AP activates once; then 3300 mV quiet. 0/5/9 V no trip; 10 V 0 mV + trigger
Bench supply: output OFF (V at terminals)	-0.125 V
Bench supply: output ON @ 0 V (V at terminals)	0.04 V

Bench supply: output ON @ 12 V (V at terminals)	12.08 V

Retest status: 2026-09-03 Phase 1 (no 680): PIA swapped tip=black/pin1, sleeve=red/pin2. B-1 PASS. B-2 sustained PASS; connection transient FAIL OPEN. Dry-contact 10/10. Rev 2 12/24 bench PASS. Rev 1 swapped PIA: Relay Contact PASS (24 V from Spark+ while in Relay Contact also triggered); 12/24 idle+fire PASS; Spark+ output idle+fire PASS. Pilot NOT TESTED. Overall OPEN. See AP-VVR-005.

Revision History

Rev	Date	Description
A	2026-08-25	Procedure opened; fillable form in inventory app
A	2026-09-01	Issued — Rev 2 bench validation; Gate A FAIL